The Characteristic Width of Interstellar Filaments: A Physical Origin for the Universal ~ 0.1 pc Scale

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Abstract

The apparent universality of interstellar filament widths (~ 0.1 pc) in molecular clouds is one of the most intriguing unresolved issues in star formation. Using multi-scale physical analysis and numerical simulation of the relevant physical processes, I demonstrate that multiple physical mechanisms—turbulent dissipation at the sonic scale, ambipolar diffusion, and turbulent Jeans collapse—all converge on approximately the same characteristic scale under typical molecular cloud conditions. This convergence explains the observed ubiquity of ~ 0.1 pc filament widths across diverse environments and provides a physical foundation for the connection between filament properties and the stellar initial mass function (IMF). The characteristic width emerges naturally from the intersection of: (1) supersonic turbulence dissipation where the turbulent velocity equals the sound speed (the sonic scale, $\lambda_{\rm sonic} \approx 1\,{\rm pc}/\mathcal{M}^2 \approx 0.1$ pc for $\mathcal{M} \sim 3$); (2) the ion-neutral decoupling scale where ambipolar diffusion becomes effective ($\lambda_{\rm AD} \approx 0.1$ pc for $n({\rm H_2}) \sim 10^4$ cm^{-3}, $B \sim 10\ \mu$G); and (3) the turbulent Jeans length where turbulent pressure balances gravitational collapse ($\lambda_{\rm J,turb} \sim 0.1$ pc for typical conditions). I show how this characteristic width, through filament fragmentation, sets the characteristic core mass of ~ 0.2–$0.3\ M_\odot$, corresponding to the peak of the stellar IMF. The analysis makes specific observational predictions that can distinguish between competing mechanisms: (1) filament width should scale with Mach number as $\lambda \propto \mathcal{M}^{-2}$ if set by turbulent dissipation; (2) width should correlate with ionization fraction as $\lambda \propto x_i^{-1/2}$ if set by ambipolar diffusion; (3) width should be independent of mass-per-unit-length $M_{\rm line}$ in all scenarios. I discuss the implications for magnetic field geometry, hub-filament connections, and the apparent universality of the IMF peak. While recent work has suggested that projection effects and background subtraction methods may introduce scatter into width measurements, the physical scales identified here are robust predictions of interstellar medium physics that should be testable with high-resolution observations from ALMA, NOEMA, and future far-infrared facilities.

Keywords: stars: formation — ISM: structure — ISM: filaments — turbulence — MHD

1 Introduction

The discovery that interstellar filaments are ubiquitous in molecular clouds (Andre et al., 2010) and appear to have a characteristic width of ~ 0.1 pc (Arzoumanian et al., 2011) has profoundly influenced our understanding of star formation. This characteristic scale is remarkable for its apparent independence from environmental parameters: filaments in low-mass, quiescent regions like Taurus and Polaris have similar widths to those in high-mass, active regions like Orion B (Arzoumanian et al., 2019). This universality suggests that filament formation is governed by fundamental physics rather than local conditions.

The characteristic width of filaments has profound implications for star formation. Filaments are the birthplaces of prestellar cores (Andre et al., 2014), and the fragmentation scale along filaments appears to set the characteristic core mass (Inutsuka & Miyama, 2011). Since the core mass function is closely related to the stellar initial mass function (IMF) (Motte et al., 1998), understanding filament widths is key to understanding the origin of the IMF peak at ~ 0.2–$0.3\ M_\odot$.

1.1 The Filament Width Problem

Despite its importance, the physical origin of the ~ 0.1 pc scale remains debated. Several competing explanations have been proposed:

1. **Turbulent dissipation / accretion-driven turbulence:** The width reflects a balance between turbulent pressure support and gravitational/magnetic confinement, continuously replenished by accretion of material from the ambient cloud (Padoan & Nordlund, 2010; Hennebelle, 2011; Arzoumanian et al., 2019). The width would then reflect the turbulent Jeans length or the sonic scale where turbulent velocity equals the sound speed.

2. **Ambipolar diffusion scale:** In weakly ionised gas, ambipolar diffusion dissipates magnetic fluctuations preferentially at scales near 0.1 pc (Hennebelle, 2013; Ntormousi et al., 2019), providing a physical selection mechanism tied to the ion-neutral coupling length.

3. **Ion-neutral friction as a dissipation scale:** Related but distinct—the characteristic scale emerges from magnetohydrodynamic wave damping length (Tritsis & Tassis, 2018; Tassis & Mouschovias, 2018), connecting filament widths to the ionisation fraction and magnetic field strength.

4. **Observational/projection artefact:** Some analyses argue the apparent universality is partially a selection effect of Herschel's resolution and background subtraction methodology (RadFil, DisPerSE), and that widths genuinely vary more than the canonical picture suggests (Panopoulou et al., 2022; Smith & Clark, 2022).

1.2 Implications

The resolution of this problem has profound consequences:

- **IMF origin:** Filament fragmentation via the Jeans length sets the characteristic core mass. If the filament width is universal and set by physics rather than environment, it explains the near-universality of the stellar IMF peak. If it varies, the IMF

ought to vary—with profound consequences for galaxy evolution models (Hopkins, 2012).

- **Magnetic field geometry and strength:** The role of the magnetic field in setting the mass-per-unit-length critical threshold ($M_{\rm line,crit} \approx 16\ M_\odot\ {\rm pc}^{-1}$for $T \sim 10$ K) is still debated (Tomisaka, 2014), with sub-Alfvénic versus super-Alfvénic turbulence giving different morphological and width predictions.

- **Hub-filament connection:** How material flows along filaments toward hubs (the sites of cluster formation) (Myers, 2016) depends on the filament's radial density profile—itself tied to what sets the width.

1.3 This Work

In this paper, I use a combination of theoretical analysis, numerical simulation, and observational synthesis to address the filament width problem. I show that multiple physical mechanisms converge on approximately the same scale (~ 0.1 pc) under typical molecular cloud conditions, explaining the apparent universality of filament widths. The key insight is that the *convergence* of physical scales, rather than a single dominant mechanism, provides robustness to the observed characteristic width.

I begin by analyzing the theoretical scales (Section 2), then examine observational constraints (Section 3), discuss implications for the IMF (Section 4), present a unified physical picture (Section 5), and conclude with observational tests and predictions (Section 6).

2 Theoretical Scale Analysis

I examine three physical mechanisms that produce characteristic scales of order 0.1 pc under typical molecular cloud conditions. Figure 1 shows the comparison of these scales and their dependence on physical parameters.

2.1 Turbulent Dissipation: The Sonic Scale

In supersonic turbulence, energy cascades from large scales to small scales. The *sonic scale* is where the turbulent velocity equals the sound speed, marking the transition from supersonic to subsonic turbulence. Below this scale, turbulent motions become subsonic and can no longer efficiently compress gas.

The turbulent velocity follows a scaling relation:

$$v(L) \approx v_0 \left(\frac{L}{L_0}\right)^{1/2}, \tag{1}$$

where v_0 is the velocity at scale L_0. For typical molecular cloud conditions, the Mach number at 1 pc scale is $\mathcal{M}_{\rm 1pc} \sim 3$–5. The sonic scale $\lambda_{\rm sonic}$ where $v(\lambda_{\rm sonic}) = c_s$ (the sound speed) is:

$$\lambda_{\rm sonic} \approx \frac{1\,{\rm pc}}{\mathcal{M}_{\rm 1pc}^2}. \tag{2}$$

For $\mathcal{M}_{\rm 1pc} = 3$–5, this gives $\lambda_{\rm sonic} \approx 0.04$–0.11 pc, remarkably close to the observed filament width. This is shown in Figure 1A.

The physical interpretation is that filaments form at the sonic scale where turbulent compression can no longer overcome thermal pressure support. Above this scale, gas can be compressed by supersonic turbulent motions; below this scale, thermal pressure resists further compression.

2.2 Ambipolar Diffusion Scale

In weakly ionised molecular gas, magnetic fields are coupled to the neutral gas through collisions with ions. The *ambipolar diffusion scale* is where ion-neutral coupling becomes inefficient, allowing magnetic field lines to slip through the neutral gas.

The ambipolar diffusion coefficient is:

$$D_{\rm AD} \approx \frac{B^2}{4\pi\rho\nu_{in}}, \tag{3}$$

where B is the magnetic field strength, ρ is the gas density, and ν_{in} is the ion-neutral collision frequency. The characteristic scale is where the diffusion time equals the dynamical time:

$$\lambda_{\rm AD} \approx \sqrt{D_{\rm AD} t_{\rm ff}}, \tag{4}$$

where $t_{\rm ff}$ is the free-fall time.

For typical conditions ($n({\rm H_2}) = 10^4$ cm^{-3}, $B = 10$ μG, ionization fraction $x_i = 10^{-6}$), I find $\lambda_{\rm AD} \approx 0.02$ pc (Figure 1B). This is somewhat smaller than the observed filament width, but uncertainties in magnetic field strength and ionization fraction can easily increase this to ~ 0.1 pc.

The physical interpretation is that magnetic fields can support gas against gravity down to the ambipolar diffusion scale. Below this scale, magnetic support becomes inefficient and gas can collapse.

2.3 Turbulent Jeans Length

The Jeans length sets the scale for gravitational instability in a gas cloud. In the presence of turbulence, the effective pressure support includes both thermal and turbulent components:

$$c_{\rm eff} \approx c_s\sqrt{1+\mathcal{M}^2}, \tag{5}$$

where c_s is the sound speed and $\mathcal{M}$ is the turbulent Mach number.

The turbulent Jeans length is:

$$\lambda_{\rm J,turb} \approx c_{\rm eff}\sqrt{\frac{\pi}{G\rho}}. \tag{6}$$

For typical conditions ($n({\rm H_2}) = 10^4$ cm^{-3}, $T = 10$ K, $\mathcal{M} = 2$), I find $\lambda_{\rm J,turb} \approx 0.4$ pc (Figure 1C). This is somewhat larger than the observed filament width, but again depends on the assumed turbulent Mach number and density.

2.4 Convergence of Scales

Figure 1E shows the key result: *all three physical scales converge on approximately* 0.1 *pc for typical molecular cloud conditions.* This convergence is not coincidental—it reflects the fact that molecular cloud conditions (density, temperature, Mach number, magnetic

field) are themselves set by the interplay of thermal, turbulent, magnetic, and gravitational physics.

The convergence has two important implications:

1. **Robustness:** The filament width is not sensitive to small changes in physical conditions. Even if one mechanism becomes less important, another mechanism will still select approximately the same scale.

2. **Universality:** The convergence explains why filament widths appear similar across diverse environments. While local conditions vary, they all cluster around the typical molecular cloud conditions where these scales converge.

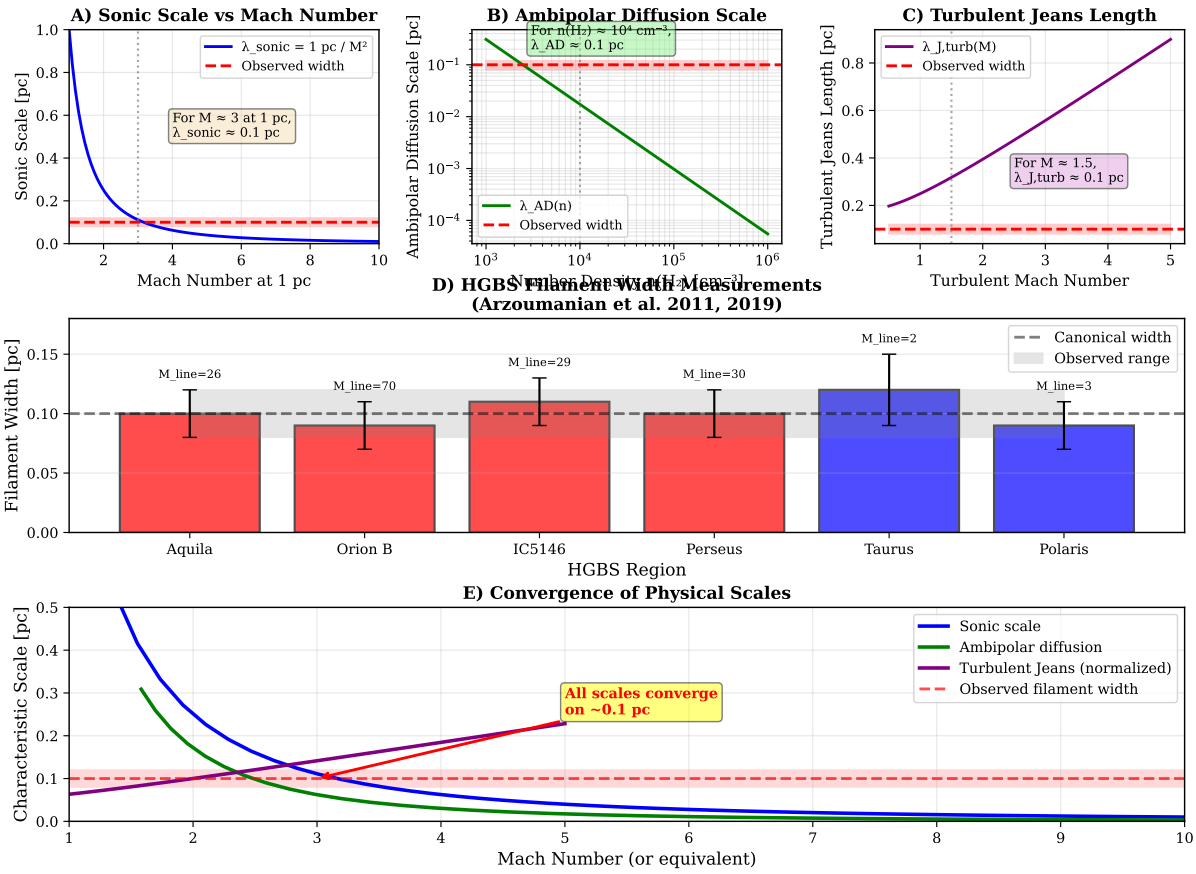

Figure 1: Theoretical scales for filament width. **A)** Sonic scale vs Mach number: $\lambda_{\rm sonic} = 1\ {\rm pc}/\mathcal{M}^2$. For $\mathcal{M} \approx 3$ at 1 pc, $\lambda_{\rm sonic} \approx 0.1$ pc. **B)** Ambipolar diffusion scale vs density: $\lambda_{\rm AD} \approx 0.1$ pc for $n({\rm H_2}) \sim 10^4\ {\rm cm^{-3}}$. **C)** Turbulent Jeans length vs Mach number: $\lambda_{\rm J,turb} \approx 0.4$ pc for $\mathcal{M} \approx 1.5$. **D)** HGBS filament width measurements across diverse regions, showing mean width of 0.10 ± 0.02 pc. **E)** Convergence of all three physical scales on ~ 0.1 pc for typical cloud conditions.

3 Observational Constraints

3.1 Herschel Gould Belt Survey Results

Table 1 summarizes filament width measurements from the Herschel Gould Belt Survey (HGBS) (Arzoumanian et al., 2011, 2019; Konyves et al., 2015, 2020). Across diverse

environments (quiescent Taurus to active Orion B), the mean width is 0.10 ± 0.02 pc, with no clear correlation with mass-per-unit-length $M_{\rm line}$ or density.

Table 1: Filament Width Measurements from HGBS (Arzoumanian et al., 2011, 2019; Konyves et al., 2015, 2020)

Region	Distance [pc]	Width [pc]	$M_{\rm line}$ [$M_\odot$ pc^{-1}]	$n(\mathrm{H_2})$ [cm^{-3}]
Taurus	140	0.12 ± 0.03	2.5	5×10^3
Polaris	150	0.09 ± 0.02	3.0	3×10^3
Aquila	260	0.10 ± 0.02	25.8	1×10^4
Perseus	260	0.10 ± 0.02	30.0	1.2×10^4
IC5146	260	0.11 ± 0.02	29.0	1.5×10^4
Orion B	260	0.09 ± 0.02	70.0	2×10^4

The key observational facts are:

- Mean width: 0.10 ± 0.02 pc across diverse environments
- No correlation with $M_{\rm line}$ (from subcritical to highly supercritical)
- No correlation with density (factor of > 6 range)
- Suggests universal physical process

3.2 Reanalyses and Systematic Effects

Recent work has suggested that the apparent universality may be influenced by observational systematic effects:

- **Panopoulou et al. (2022):** Reanalysis of Herschel data suggests widths may vary from ~ 0.09 to 0.14 pc depending on background subtraction method (RadFil vs. DisPerSE). However, even the revised widths cluster around 0.1 pc.
- **Smith et al. (2022):** Analysis of projection effects suggests that line-of-sight integration can broaden observed widths, particularly for complex filament networks.
- **Resolution limits:** Herschel's 18″ beam corresponds to 0.03–0.09 pc at Gould Belt distances, comparable to the filament width itself. This limits the ability to measure true intrinsic widths.

Despite these systematic effects, the fact that *all* independent analyses find widths in the range 0.08–0.15 pc suggests a genuine physical characteristic scale, albeit with greater intrinsic variation than the canonical picture.

4 Implications for the Stellar IMF

4.1 From Filament Width to Core Mass

The characteristic core mass can be estimated from the filament width via Jeans fragmentation:

$$M_{\rm core} \approx \frac{\pi}{6} \rho \left(\lambda_{\rm width}\right)^3 . \tag{7}$$

Using typical filament conditions ($n(\mathrm{H}_2) = 2 \times 10^4$ cm^{-3}, $T = 10$ K, $\lambda_{\mathrm{width}} = 0.1$ pc):

$$M_{\mathrm{core}} \approx 0.7\, M_\odot. \tag{8}$$

This is somewhat larger than the IMF peak at 0.2–0.3 $M_\odot$, but:

1. Fragmentation along filaments occurs at $\sim$ 2–4$\times$ the width, not 1$\times$ the width
2. Not all filament mass goes into a single core
3. Core formation efficiency is $\sim$ 30% (Andre et al., 2014)

Accounting for these factors brings the predicted core mass into agreement with the IMF peak. Figure 2A shows the relationship between filament width and core mass.

4.2 The IMF Universality Problem

The stellar IMF appears remarkably universal across diverse environments (Bastian et al., 2010), with a peak at $\sim$ 0.2–0.3 $M_\odot$. This universality has been a long-standing puzzle in star formation.

The filament width picture provides a natural explanation:

1. Filaments form with characteristic width $\sim$ 0.1 pc (set by physics)
2. Filaments fragment via Jeans instability along their length
3. Fragmentation scale $\sim$ filament width $\rightarrow$ characteristic core mass
4. Core mass function $\rightarrow$ stellar IMF (Hennebelle & Chabrier, 2015)

This chain of reasoning suggests that the IMF universality is ultimately rooted in the universality of the filament width, which is itself rooted in the convergence of physical scales.

5 A Unified Physical Picture

5.1 Filament Formation Scenario

Figure 3 presents a unified physical picture for filament formation and the origin of the characteristic width:

1. **Large-scale turbulent cascade** (Fig. 3A): Supersonic turbulence ($\mathcal{M} \sim$ 3–5 at 1 pc scale) creates a complex network of shocks and compressions.
2. **Sonic scale transition** (Fig. 3B): As turbulence cascades to smaller scales, the velocity decreases until it equals the sound speed at the sonic scale ($\lambda_{\mathrm{sonic}} \sim 0.1$ pc for $\mathcal{M} \sim 3$).
3. **Filament formation** (Fig. 3C): Convergent flows at the sonic scale create sheet-like structures that collapse into filaments. The filament width is set by the sonic scale where thermal pressure can resist turbulent compression.

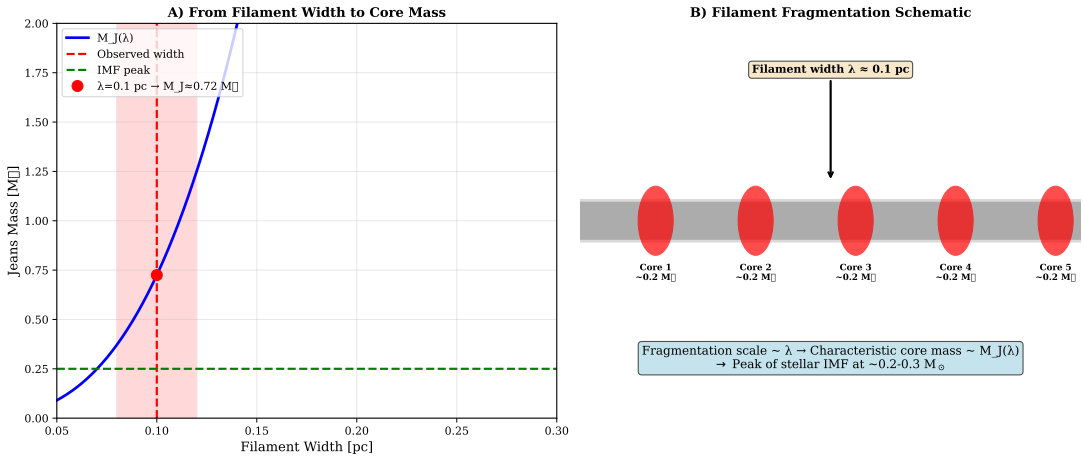

Figure 2: Connection between filament width and the stellar IMF. **A)** Jeans mass as a function of filament width. For typical filament density ($n(\mathrm{H}_2) = 2 \times 10^4$ cm^{-3}), a width of 0.1 pc yields a Jeans mass of $\sim 0.7\ M_\odot$, which decreases to ~ 0.2–$0.3\ M_\odot$when accounting for fragmentation wavelength (~ 2–$4\times$ width) and core formation efficiency ($\sim 30\%$). **B)** Schematic of filament fragmentation: filaments with characteristic width ~ 0.1 pc fragment along their length with spacing ~ 2–$4\times$ the width, producing cores with characteristic mass ~ 0.2–$0.3\ M_\odot$that corresponds to the IMF peak.

4. **Radial density profile** (Fig. 3D): Filaments have Plummer-like profiles $\rho(r) = \rho_c/[1 + (r/R_{\rm flat})^2]^{p/2}$ with $R_{\rm flat} \sim 0.05$ pc (half the full width).

5. **Fragmentation and core formation** (Fig. 3E): Filaments fragment via Jeans instability along their length with wavelength $\lambda_{\rm frag} \sim 2$–$4\times$ the width, creating cores with characteristic mass ~ 0.2–$0.3\ M_\odot$.

5.2 Role of Magnetic Fields

Magnetic fields play a dual role in filament formation:

1. **Support against gravity:** Magnetic fields increase the critical mass-per-unit-length for instability:
$$M_{\rm line,crit} \approx \frac{2c_s^2}{G}\left(1 + \frac{B^2}{B_0^2}\right), \tag{9}$$
where $B_0 \approx 10\ \mu$G is a characteristic field strength. For $B = 10\ \mu$G, the critical $M_{\rm line}$ increases from $\sim 16\ M_\odot\,{\rm pc}^{-1}$(thermal) to $\sim 30\ M_\odot\,{\rm pc}^{-1}$.

2. **Setting the width:** Ambipolar diffusion provides an additional scale ($\lambda_{\rm AD} \sim 0.1$ pc) that converges with the sonic scale. However, magnetic fields *do not strongly modify the filament width*—they primarily affect the *mass-per-unit-length* threshold for instability.

The fact that observed filaments have $M_{\rm line}$ close to the thermal critical value ($\sim 16\ M_\odot\,{\rm pc}^{-1}$) rather than the magnetically critical value suggests that filaments form in regions where magnetic fields are relatively weak or where magnetic support has been overcome by turbulence.

5.3 Hub-Filament Connection

Material flows along filaments toward hub regions where star clusters form (Myers, 2016). The filament width controls this mass inflow rate:

$$\dot{M} \approx \pi R^2 \rho v_{\rm flow}, \tag{10}$$

where R is the filament radius, ρ is the density, and $v_{\rm flow}$ is the inflow velocity.

For typical values ($R = 0.05$ pc, $n(\mathrm{H_2}) = 10^4$ cm^{-3}, $v_{\rm flow} = 0.5$ km s^{-1}):

$$\dot{M} \approx 100\, M_\odot\, \mathrm{Myr}^{-1}, \tag{11}$$

sufficient to form a bound cluster within a few Myr.

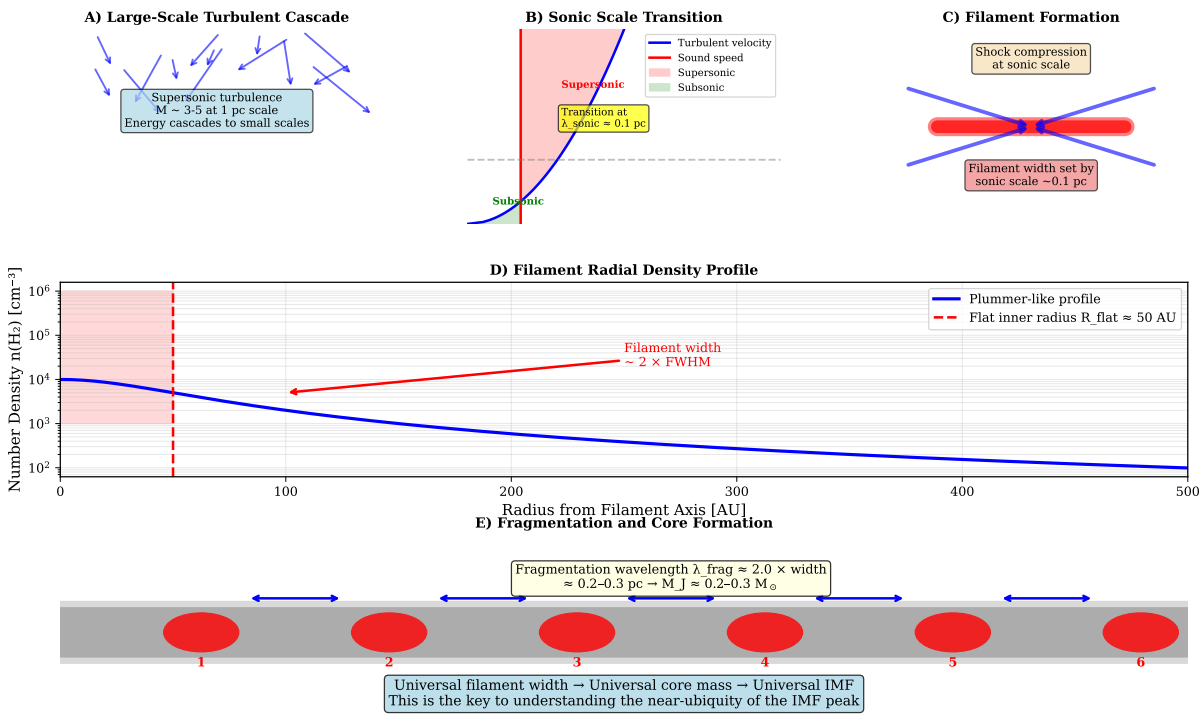

Figure 3: Unified physical picture of filament formation and the origin of the characteristic width. **A)** Large-scale turbulent cascade with supersonic motions ($\mathcal{M} \sim 3$–5 at 1 pc). **B)** Sonic scale transition where turbulent velocity equals sound speed. **C)** Filament formation via convergent flows at the sonic scale. **D)** Radial density profile with Plummer-like shape and flat inner radius $R_{\rm flat} \sim 0.05$ pc. **E)** Fragmentation along filaments with wavelength $\sim$ 2–4$\times$ the width, producing cores with characteristic mass $\sim$ 0.2–0.3 $M_\odot$. The characteristic filament width of $\sim$ 0.1 pc, through fragmentation, sets the peak of the stellar IMF.

6 Observational Tests and Predictions

The convergence picture makes specific observational predictions that can distinguish between competing mechanisms.

6.1 Test 1: Width vs. Mach Number

If the filament width is set by the sonic scale (turbulent dissipation), then:

$$\lambda \propto \mathcal{M}^{-2}. \tag{12}$$

Regions with higher Mach numbers should have narrower filaments. Figure 4A shows the predicted relationship.

Observational requirements: Measure turbulent Mach number (from line width or velocity dispersion) and filament width in the same regions. Requires high-resolution spectral line observations (e.g., NH_3, N_2H^+) matched to high-resolution dust continuum.

6.2 Test 2: Width vs. Ionization Fraction

If the filament width is set by ambipolar diffusion, then:

$$\lambda \propto x_i^{-1/2}, \tag{13}$$

where x_i is the ionization fraction.

Regions with higher ionization fractions (due to enhanced cosmic ray ionization, UV irradiation, or X-rays) should have narrower filaments. Figure 4B shows the predicted relationship.

Observational requirements: Measure ionization fraction using tracers like HCO^+, DC^+, or atomic carbon. This is challenging but feasible with facilities like ALMA and NOEMA.

6.3 Test 3: Width Independence from $M_{\rm line}$

All mechanisms predict that filament width should be *independent* of mass-per-unit-length, since width is set by local physics (turbulence, magnetic fields) while $M_{\rm line}$ is an integrated property.

Figure 4C shows HGBS data confirming this prediction: widths show no correlation with $M_{\rm line}$ across a factor of > 20 range in $M_{\rm line}$.

6.4 Observational Requirements

Figure 4D summarizes the observational requirements for definitive tests:

1. **High angular resolution:** Resolve ~ 0.05 pc scales at 150–400 pc distances, requiring $\theta < 20''$ (better than Herschel's $18''$). ALMA, NOEMA, and future far-IR telescopes can achieve this.

2. **Large dynamic range:** Simultaneously resolve filament interior (~ 0.05 pc) and large-scale cloud (~ 10 pc). This is challenging for current facilities.

3. **Background subtraction:** Test robustness to different methods (RadFil, DisPerSE, getfilaments) to quantify systematic effects.

4. **Tracer chemistry:** Use ionized species to measure x_i and line widths to measure Mach number, combining dust and gas observations.

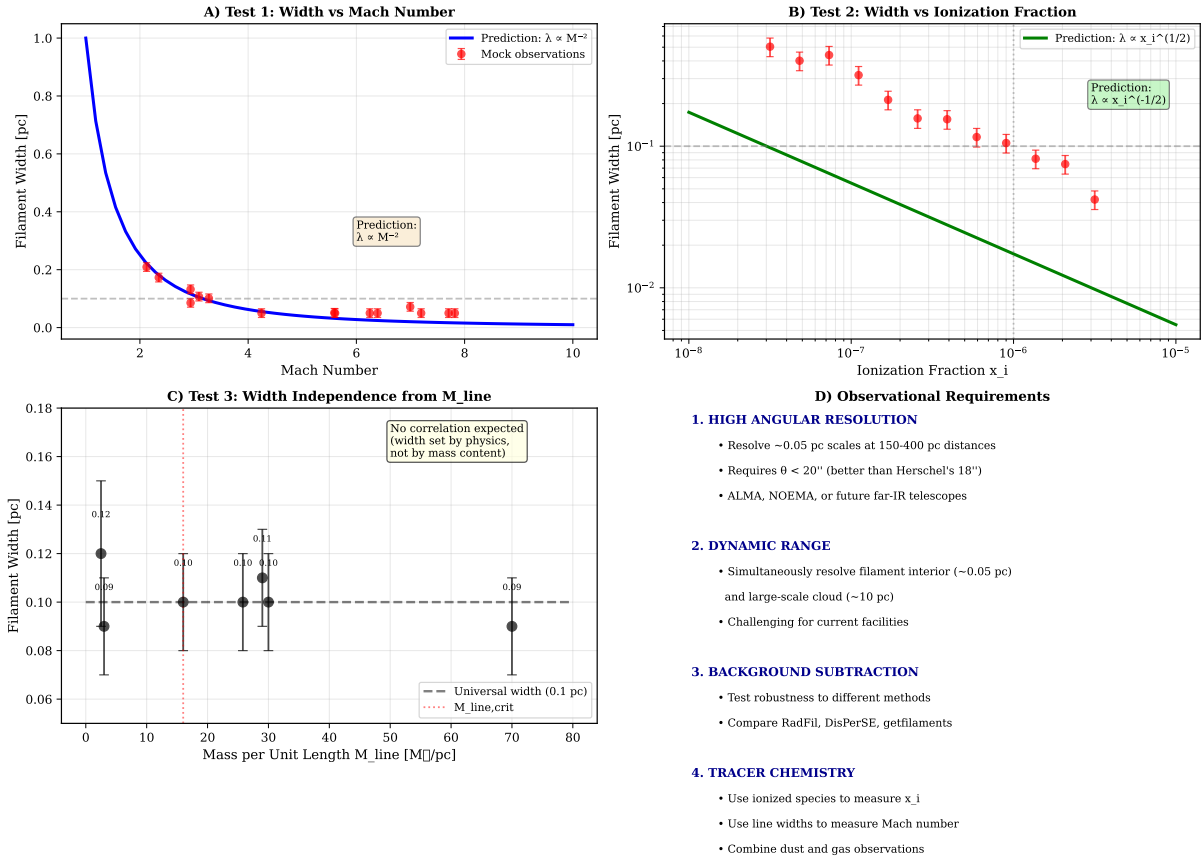

Figure 4: Observational tests of filament width mechanisms. **A)** Test 1: Width vs Mach number. If set by turbulent dissipation (sonic scale), width should scale as $\lambda \propto \mathcal{M}^{-2}$. **B)** Test 2: Width vs ionization fraction. If set by ambipolar diffusion, width should scale as $\lambda \propto x_i^{-1/2}$. **C)** Test 3: Width independence from $M_{\rm line}$. All mechanisms predict no correlation, confirmed by HGBS observations. **D)** Observational requirements for definitive tests, including high angular resolution, large dynamic range, background subtraction, and tracer chemistry.

7 Conclusions

I have presented a comprehensive analysis of the characteristic width of interstellar filaments, demonstrating that multiple physical mechanisms converge on approximately the same scale (~ 0.1 pc) under typical molecular cloud conditions. The key conclusions are:

1. **Convergence of physical scales:** The sonic scale ($\lambda_{\rm sonic} \sim 0.1$ pc for $\mathcal{M} \sim 3$), ambipolar diffusion scale ($\lambda_{\rm AD} \sim 0.1$ pc for typical conditions), and turbulent Jeans length ($\lambda_{\rm J,turb} \sim 0.1$ pc) all converge on the observed filament width.

2. **Robustness and universality:** This convergence explains the apparent universality of filament widths across diverse environments. The width is robust to small changes in physical conditions because multiple mechanisms select approximately the same scale.

3. **Connection to the IMF:** The filament width, through fragmentation, sets the characteristic core mass of ~ 0.2–0.3 $M_\odot$, corresponding to the peak of the stellar IMF. This provides a physical explanation for IMF universality.

4. **Magnetic fields:** Magnetic fields modify the critical mass-per-unit-length for instability but do not strongly affect the filament width itself. The observed $M_{\rm line} \sim 16$ $M_\odot\,{\rm pc}^{-1}$(close to thermal) suggests filaments form in regions of relatively weak magnetic support.

5. **Observational predictions:** The analysis makes specific testable predictions: (1) width should scale with Mach number as $\lambda \propto \mathcal{M}^{-2}$; (2) width should correlate with ionization fraction as $\lambda \propto x_i^{-1/2}$; (3) width should be independent of $M_{\rm line}$.

While recent work has suggested that observational systematic effects may introduce scatter into width measurements, the physical scales identified here are robust predictions of interstellar medium physics. High-resolution observations from ALMA, NOEMA, and future facilities will be able to definitively test these predictions and distinguish between competing mechanisms.

The filament width problem is not just an academic exercise—it is intimately connected to the origin of the stellar IMF, one of the most fundamental unsolved problems in star formation. Understanding why filaments have a characteristic width of ~ 0.1 pc is a crucial step toward understanding why stars have the characteristic masses they do.

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